

Open Policy Analysis

Proposal for a course in a Master of Public Policy program

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Motivation

Policy analysis is one of the main channels that connect evidence with policy. In practice, the credibility of a policy analysis is judged much more by the reputation or ideology of its authors than by its analytical quality. A key reason is that most analyses cannot be inspected. The data, code, and assumptions behind a policy estimate are not available, and the final number carries degrees of freedom that a reader cannot see. Public debates then feature advocates and opponents bringing reports that seem to come from parallel universes.

Research has faced the same problem and built a response. The open science movement raised the credibility of research by making data, code, and pre-registered hypotheses public. Open Policy Analysis (OPA) brings those practices into policy analysis. The framework sets out three principles (open output, open analysis, and open materials) and applies them to four layers of a report, each written for a different reader, from the policymaker who sees one pre-specified result and how it moves with the key assumptions, down to the analyst who can reproduce everything.

Two things have changed since this course was first proposed in 2022. Agentic AI tools can now build a large part of an open analysis in days rather than months, so the cost that kept OPA from becoming routine practice is falling. The same tools make it easy to produce an analysis that looks finished and is wrong. The course therefore trains students to build open analyses with these tools and to verify every step.

Aim

The course trains the next generation of policy analysts to conduct transparent, reproducible, and verifiable policy analyses, and to use agentic AI tools without losing the understanding that makes an analysis theirs.

Content

The course has two parts: concepts and applications.

In the conceptual part, students study the evidence-to-policy chain and why the credibility of a policy analysis is judged by proxies. They then review the open science response to the reproducibility crisis in research and the OPA framework that translates it into policy analysis.

In the applied part, students learn the tools of open policy analysis: version control (Git), open-source programming for data analysis (R or Python), dynamic documents (Quarto), interactive tools (Shiny), and agentic coding harnesses such as Claude Code. Two sessions cover the concepts a student needs before using an agentic harness: version control, Markdown documentation, verification plans, token management, skills, hooks, and global and local memory. Students then review two high-profile policy analyses and study how to open them. Finally, each student builds an open analysis of a live policy question, with AI tools and with every step verified. Students disclose AI use for each deliverable, stating who conceived it, who executed it, and who checked it. They leave with a portfolio piece that any reader can inspect and rerun.

Relationship to other courses

The course complements courses in data science for public policy. Its orientation is toward policy analysis in practice, with less emphasis on specific statistical methods, and its tools are chosen for the format of an analysis rather than for any one estimation technique. Open policy analysis can also be taught as a three- to four-week module inside a traditional policy analysis course, covering

the framework, the tools, and a short build.

Structure and main readings

For a semester of 13 teaching weeks, the proposed structure is:

1. **Concepts for Open Policy Analysis** (4 weeks)
 - a. *Motivation* (2 weeks). The evidence-to-policy chain, dueling certitudes, and the parallel between the reproducibility crisis in science and the low credibility of policy analysis. Readings: Manski (2013); Christensen, Freese, and Miguel (2019); Frankfurt (2005).
 - b. *Does policy analysis matter?* (1 week). What is known about whether, and when, policy analysis improves policy decisions, including its role in bringing evidence and transparency into decision processes. Readings: Friedman (2017).
 - c. *Framework* (1 week). The OPA framework and its connection to initiatives that raise the rigor of policy analysis and policy prediction. Readings: Hoces de la Guardia, Grant, and Miguel (2021); Sunstein (2013); Tetlock and Gardner (2016).
2. **OPA in practice** (9 weeks)
 - a. *Tools* (4 weeks). Git, R or Python, Quarto, Shiny, and agentic coding harnesses, with two sessions on the concepts needed to use a harness and one on verification plans. Readings: Bryan (2018); Xie, Allaire, and Golemund (2018); Wickham (2021); Tao (2026).
 - b. *Examples* (2 weeks). In-depth discussion of two high-profile policy analyses and of one open analysis built with agentic tools. Readings: Saez and Zucman (2019); Congressional Budget Office (2019); Hoces de la Guardia (2026).
 - c. *Build your own OPA* (3 weeks). Development, verification, disclosure, and presentation of the students' open analyses. No readings.

Readings

- Bryan, Jennifer. *Happy Git and GitHub for the useR*. 2018. <https://happygitwithr.com>
- Christensen, Garret, Jeremy Freese, and Edward Miguel. *Transparent and Reproducible Social Science Research*. University of California Press, 2019.
- Congressional Budget Office. *The Effects on Employment and Family Income of Increasing the Federal Minimum Wage*. 2019. <https://www.cbo.gov/publication/55410>
- Frankfurt, Harry G. *On Bullshit*. Princeton University Press, 2005.
- Friedman, Lee S., ed. *Does Policy Analysis Matter? Exploring Its Effectiveness in Theory and Practice*. University of California Press, 2017.
- Hoces de la Guardia, Fernando, Sean Grant, and Edward Miguel. "A framework for open policy analysis." *Science and Public Policy*, 2021.
- Hoces de la Guardia, Fernando. *An open policy analysis of the economic effects of AI*. 2026. <https://fhoces.github.io/opa-ai-macro-econ-scenarios/>
- Kahneman, Daniel. *Thinking, Fast and Slow*. Macmillan, 2011.
- Manski, Charles F. *Public Policy in an Uncertain World: Analysis and Decisions*. Harvard University Press, 2013.
- Saez, Emmanuel, and Gabriel Zucman. Letter to Senator Elizabeth Warren on a progressive wealth tax. January 18, 2019.
- Sunstein, Cass R. *Simpler: The Future of Government*. Simon and Schuster, 2013.
- Tao, Terence. "Mathematics in the age of AI." arXiv:2608.16753, 2026.
- Tetlock, Philip E., and Dan Gardner. *Superforecasting: The Art and Science of Prediction*. Random House, 2016.
- Watts, Duncan J. *Everything Is Obvious: Once You Know the Answer*. Crown Business, 2011.
- Wickham, Hadley. *Mastering Shiny*. O'Reilly Media, 2021.
- Xie, Yihui, J. J. Allaire, and Garrett Golemund. *R Markdown: The Definitive Guide*. Chapman and Hall/CRC, 2018.